

## SAFETY DATA SHEET

Creation Date 13-April-2010

Revision Date 26-March-2024

Revision Number 3

### 1. Identification

**Product Name** [(R)-Binap RuCl p-cymene]Cl  
**Cat No. :** Z33436  
**CAS-No** 145926-28-9  
**Synonyms** No information available  
**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                                |            |
|--------------------------------------------------------------------------------|------------|
| <b>Acute oral toxicity</b>                                                     | Category 4 |
| <b>Acute dermal toxicity</b>                                                   | Category 4 |
| <b>Acute Inhalation Toxicity</b>                                               | Category 4 |
| <b>Skin Corrosion/Irritation</b>                                               | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                                       | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b>                        | Category 2 |
| Target Organs - Respiratory system, Optic nerve, Central nervous system (CNS). |            |
| <b>Specific target organ toxicity - (repeated exposure)</b>                    | Category 2 |
| Target Organs - Kidney, Liver.                                                 |            |

#### Label Elements

##### **Signal Word**

Danger

**Hazard Statements**

Harmful if swallowed, in contact with skin or if inhaled  
Causes skin irritation  
Causes serious eye irritation  
May cause respiratory irritation  
Causes damage to organs  
May cause damage to organs  
May cause damage to organs through prolonged or repeated exposure  
Harmful if inhaled

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Do not get in eyes, on skin, or on clothing  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

IF ON SKIN: Wash with plenty of soap and water  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
IF exposed or concerned: Get medical advice/attention  
Call a POISON CENTER/ doctor if you feel unwell  
Rinse mouth  
Take off contaminated clothing and wash it before reuse

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Poison, may be fatal or cause blindness if swallowed

### 3. Composition/Information on Ingredients

| Component                                                                                 | CAS-No      | Weight % |
|-------------------------------------------------------------------------------------------|-------------|----------|
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1-binaphthyl](p-cymene)ruthenium(II) chloride | 145926-28-9 | 95       |
| Methanol                                                                                  | 67-56-1     | 5        |

### 4. First-aid measures

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention.

|                                                               |                                                                                                                                                                                                                                                                                                              |
|---------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                                             | Remove to fresh air. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Get medical attention. If not breathing, give artificial respiration. |
| <b>Ingestion</b>                                              | Do NOT induce vomiting. Get medical attention.                                                                                                                                                                                                                                                               |
| <b>Most important symptoms/effects<br/>Notes to Physician</b> | No information available.<br>Treat symptomatically                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

|                                         |                                                                                       |
|-----------------------------------------|---------------------------------------------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water spray, carbon dioxide (CO <sub>2</sub> ), dry chemical, alcohol-resistant foam. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                                                              |
| <b>Flash Point</b>                      | No information available                                                              |
| <b>Method -</b>                         | No information available                                                              |
| <b>Autoignition Temperature</b>         | No information available                                                              |
| <b>Explosion Limits</b>                 |                                                                                       |
| <b>Upper</b>                            | No data available                                                                     |
| <b>Lower</b>                            | No data available                                                                     |
| <b>Sensitivity to Mechanical Impact</b> | No information available                                                              |
| <b>Sensitivity to Static Discharge</b>  | No information available                                                              |

### Specific Hazards Arising from the Chemical

Keep product and empty container away from heat and sources of ignition.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen chloride gas.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 1                   | 1                  | N/A                     |

## 6. Accidental release measures

|                                             |                                                                                                                                       |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Avoid dust formation.                                     |
| <b>Environmental Precautions</b>            | Should not be released into the environment. See Section 12 for additional Ecological Information.                                    |
| <b>Methods for Containment and Clean Up</b> | Sweep up and shovel into suitable containers for disposal. Avoid dust formation. Use spark-proof tools and explosion-proof equipment. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                          |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Use only under a chemical fume hood. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Avoid dust formation. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place. Store under an inert atmosphere. Keep refrigerated. Incompatible Materials. Strong oxidizing agents.                                            |

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component | Alberta                                                                                            | British Columbia                      | Ontario TWAEV                         | Quebec                                                                                             | ACGIH TLV                             | OSHA PEL                                                                                                                                                                                 | NIOSH                                                                                                        |
|-----------|----------------------------------------------------------------------------------------------------|---------------------------------------|---------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| Methanol  | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | TWA: 200 ppm<br>TWA: 262 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 328 mg/m <sup>3</sup><br>Skin | TWA: 200 ppm<br>STEL: 250 ppm<br>Skin | (Vacated) TWA: 200 ppm<br>(Vacated) TWA: 260 mg/m <sup>3</sup><br>(Vacated) STEL: 250 ppm<br>(Vacated) STEL: 325 mg/m <sup>3</sup><br>Skin<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup> | IDLH: 6000 ppm<br>TWA: 200 ppm<br>TWA: 260 mg/m <sup>3</sup><br>STEL: 250 ppm<br>STEL: 325 mg/m <sup>3</sup> |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material                                      | Breakthrough time                 | Glove thickness | Glove comments         |
|-----------------------------------------------------|-----------------------------------|-----------------|------------------------|
| Nitrile rubber<br>Neoprene<br>Natural rubber<br>PVC | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                      |
|----------------------------------------|--------------------------------------|
| Physical State                         | Solid                                |
| Appearance                             | Yellow                               |
| Odor                                   | No information available             |
| Odor Threshold                         | No information available             |
| pH                                     | No information available             |
| Melting Point/Range                    | No data available                    |
| Boiling Point/Range                    | No information available             |
| Flash Point                            | No information available             |
| Evaporation Rate                       | Not applicable                       |
| Flammability (solid,gas)               | No information available             |
| Flammability or explosive limits       |                                      |
| Upper                                  | No data available                    |
| Lower                                  | No data available                    |
| Vapor Pressure                         | No information available             |
| Vapor Density                          | Not applicable                       |
| Specific Gravity                       | No information available             |
| Solubility                             | Slightly soluble in water Hydrolyses |
| Partition coefficient; n-octanol/water | No data available                    |
| Autoignition Temperature               | No information available             |
| Decomposition Temperature              | No information available             |
| Viscosity                              | Not applicable                       |
| Molecular Formula                      | C54 H46 Cl2 P2 Ru                    |
| Molecular Weight                       | 928.88                               |

## 10. Stability and reactivity

|                                  |                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                                                                 |
| Stability                        | Moisture sensitive. Air sensitive.                                                                         |
| Conditions to Avoid              | Incompatible products. Excess heat. Avoid dust formation. Exposure to moist air or water. Exposure to air. |
| Incompatible Materials           | Strong oxidizing agents                                                                                    |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen chloride gas                             |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                   |
| Hazardous Reactions              | None under normal processing.                                                                              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information Component Information

| Component | LD50 Oral                      | LD50 Dermal                   | LC50 Inhalation               |
|-----------|--------------------------------|-------------------------------|-------------------------------|
| Methanol  | LD50 = 1187 – 2769 mg/kg (Rat) | LD50 = 17100 mg/kg ( Rabbit ) | LC50 = 128.2 mg/L ( Rat ) 4 h |

Toxicologically Synergistic No information available

#### Products

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Irritating to eyes, respiratory system and skin                                          |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                                                                  | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------------------------------------------------|-------------|------------|------------|------------|------------|------------|
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1'-binaphthyl](p-cymene)ruthenium(II) chloride | 145926-28-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Methanol                                                                                   | 67-56-1     | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system Optic nerve Central nervous system (CNS)

**STOT - repeated exposure** Kidney Liver

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** No information available

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

| Component | Freshwater Algae | Freshwater Fish                            | Microtox                                                                        | Water Flea            |
|-----------|------------------|--------------------------------------------|---------------------------------------------------------------------------------|-----------------------|
| Methanol  | Not listed       | Pimephales promelas: LC50 > 10000 mg/L 96h | EC50 = 39000 mg/L 25 min<br>EC50 = 40000 mg/L 15 min<br>EC50 = 43000 mg/L 5 min | EC50 > 10000 mg/L 24h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** . Is not likely mobile in the environment.

| Component | log Pow |
|-----------|---------|
| Methanol  | -0.74   |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

| Component          | RCRA - U Series Wastes | RCRA - P Series Wastes |
|--------------------|------------------------|------------------------|
| Methanol - 67-56-1 | U154                   | -                      |

## 14. Transport information

**DOT** Not regulated  
**TDG** Not regulated  
**IATA** Not regulated  
**IMDG/IMO** Not regulated

## 15. Regulatory information

## International Inventories

| Component                                                                                  | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------------------------------------------------------|-------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1-binaphthyl](p-cymene) ruthenium(II) chloride | 145926-28-9 | -   | -    | -    | -                                             | -         | -      | -   |
| Methanol                                                                                   | 67-56-1     | X   | -    | X    | ACTIVE                                        | 200-659-6 | -      | -   |

| Component                                                                                  | CAS-No      | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------------------------------------------------|-------------|-------|----------|------|------|------|------|-------|-------|
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1-binaphthyl](p-cymene) ruthenium(II) chloride | 145926-28-9 | -     | -        | -    | -    | -    | -    | -     | -     |
| Methanol                                                                                   | 67-56-1     | X     | KE-23193 | X    | X    | X    | X    | X     | X     |

## Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component | Canada - National Pollutant Release Inventory (NPRI)                        | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|-----------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| Methanol  | Part 1, Group A Substance<br>Part 5, Individual Substances Part 4 Substance |                                                                            |                                           |

## Other International Regulations

## Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances                                                            | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Methanol  | -                                                                   | Use restricted. See item 69.<br>(see link for restriction details)<br>Use restricted. See item 75.<br>(see link for restriction details) | -                                                                                                     |

## REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                                                                 | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-------------------------------------------------------------------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1-binaphthyl](p-cymene)ruthenium(II) chloride | 145926-28-9 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Methanol                                                                                   | 67-56-1     | Listed                                                                                    | Not applicable                                                                           | Not applicable             | Not applicable                     |
|--------------------------------------------------------------------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Component                                                                                  | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
| Chloro[(R)-(+)-2,2-bis(diphenylphosphino)-1,1'-binaphthyl](p-cymene)ruthenium(II) chloride | 145926-28-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Methanol                                                                                   | 67-56-1     | 500 tonne                                                                                 | 5000 tonne                                                                               | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
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www.thermofisher.com

**Creation Date**

13-April-2010

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**Print Date**

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**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**